# Peravali Bhavana Srivalli

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 8.96 |
| Narayana Junior College | Class XII (Telangana State Board Of Intermediate Education) | 2016 | 97.50% |
| krishnaveni Talent School | Class X (Board of Secondary Education ) | 2014 | 9.80% |

## POSITIONS OF RESPONSIBILITY

Teaching assistant for Analog circuits by Dr.Abhishek kumar

Teaching assistant for Electric Circuit lab by Dr. Gajendranath chaudhury.

## RELEVANT COURSES

**Digital Domain**: Digital Electronics, Digital IC Design, Introduction to VLSI Design, Digital IC Design Lab

**Analog Domain**: Analog IC Design, Analog IC Design Lab

**Devices Courses**: Semi Conductor Device Modelling, Neuromorphic Devices for AI/ML

## SKILLS

**Coding Languages**: C,verilog, VHDL

**Tools**: Cadence Virtuoso, Xilinx Vivado, LTSpice, MATLAB, FPGA ZedBoard, Elcad, Eplan

## MTECH THESIS

**Instructor** : Dr. Abhishek Kumar

**Title :**Design of Transceiver for On board Wireless interface of distributed Control systems.

**Description:**

- The Low-Dropout regulators is developed for a wireless transceiver in cadence virtuoso (tsmc 65nm tech node ).
- The LDO regulates the input voltage of **2.5V to 1.2V** with a maximum current of **120mA**. The blocks used in the design of LDO are cascode band-gap reference, constant-gm, and operational amplifier.
- The cascode band gap reference generates a reference voltage of **1.2V** and has a high power supply rejection ratio. The constant-gm circuit gives a supply-independent bias current to the op-amps.
- The op-amps designed are folded cascode op-amps for high gain and PSRR. The design is carried out in CMOS 65nm technology.

## PROJECTS

**Real time data acquisition from ESP32 to FPGA board**

- Developed a real-time EMG data acquisition system using ESP32 and Zynq-7000 FPGA for sensor-data transfer and FPGA-side processing.
- Achieved 330 samples/s via UART and ~900 samples/s via Serial-to-Parallel transfer, with Vivado hardware configuration and Vitis software deployment.

**RTL Design for Dual Clock Asynchronous FIFO**

- Designed and implemented Dual Clock Asynchronous FIFO with different read and write clocks using RTL
- Employed Gray Encoding to avoid multi bit transistions and mitigate metastable issues.

**Implementation of 12 Hour Digital Clock Using Verilog**

- Implemented the 12 hour digital clock with AM/PM indicator using the RTL code
- Hours ,Minutes and Seconds are incremented using the counters and the time is displayed in the format HH:MM:SS

AM/PM

**8-Stage Multi-Modulus Frequency Divider (Extended Range)**

- Designed and implemented 8-stage Multi-Modulus Divider (MMD) implemented in Verilog
- The design consists of eight cascaded 2/3 divider cells with a multiplexer-based output selection scheme that extends the conventional division range from the standard band of 2 to (2.2)-1 to a continuous range of 2 to 511

## PROFESSIONAL EXPERIENCE

**Executive Design Engineer - Siemens Energy,** Aurangabad, India

- Responsible for both customer and factory engineering. Preparing and reviewing secondary control schematics and logic based on customer specification.
- Preparation of Base design including hardwired logics for controlling of On and Off operation os components of GIS SS.
- Automated report generation using Excel VBA, reducing preparation time by 50% and improving data accuracy.

## EXTRACURRICULAR

**Hobbies :** Doodling, Badminton.